

Measurement Investigation
Semester 2, 2018



Baldy's
Secondary College

Name: ANSWERS

Collaborative Component

/ 29 marks
55 MINUTES

- You will need:
- 4 sheets of paper
 - Scissors
 - Sticky tape (collect as needed)
 - Ruler

PART A

[16 marks]

1. Take one sheet of paper and join the two short edges together to make a cylinder.
2. Write "A" on this cylinder
3. Take another sheet of paper and join the two long edges together to make a cylinder.
4. Write "B" on this cylinder

Look at your cylinders. Will they both hold the same amount? Write a prediction.

A 9 B 9 Both ✓

why ✓

Complete the table below to test your prediction of the test. Give values to the nearest tenth.

Cylinder	Radius	Height	Circumference	Area of Circular Base	Volume	Area of curved surface
A	r	h	$C = 2\pi r$	$A = \pi r^2$	$V = A \times h$	$SA_c = C \times h$
	$\approx 5 \text{ cm}$	12 cm	$2\pi(5)$ $= 31.4 \text{ cm}$	$\pi(5)^2$ $= 78.5 \text{ cm}^2$	78.5×12 $= 942 \text{ cm}^3$	31.4×12 $= 376.8 \text{ cm}^2$
B	$\approx 3 \text{ cm}$	29.5 cm	$2\pi(3)$ $= 18.8 \text{ cm}$	$\pi(3)^2$ $= 28.3 \text{ cm}^2$	28.3×29.5 $= 834.9 \text{ cm}^3$	18.8×29.5 $= 554.6 \text{ cm}^2$

Conclusion

✓✓

PART B

[13 marks]

1. Take another sheet of paper. Fold it in half across the shortest side to make a long narrow rectangle and cut.
2. Stick the short ends together to make a long strip. Now stick the remaining short ends together to make a wide cylinder.
3. Write "C" on this cylinder.
4. Take another sheet of paper. Fold it in half across the shortest side to make a long narrow rectangle (again) and cut.
5. Stick the short ends together to make a long strip. Now stick the long sides together to make a long thin cylinder.
6. Write "D" on this cylinder.

Look at your four cylinders (A, B, C, D). Consider the results of the first experiment. Which will hold the most, the least? Put your cylinders in order from most to least (using their letters).
Write your prediction below.

Least _____ Most _____

A

NO MARKS

Copy the first table below and complete the 'C' and 'D'

Cylinder	Radius	Height	Circumference	Area of Circular Base	Volume	Area of curved surface
	r	h	$C = 2\pi r$	$A = \pi r^2$	$V = A \times h$	$SA_c = C \times h$
A						
B						
C	$\approx 10 \text{ cm}$	10.5 cm	$2\pi(10)$ $= 62.8 \text{ m}$	$\pi(10)^2$ $= 314.2 \text{ m}^2$	314.2×10.5 $= 3299.1 \text{ m}^3$	62.8×10.5 $= 659.4 \text{ m}^2$
D	$\approx 1.5 \text{ cm}$	5.8 cm	$2\pi(1.5)$ $= 9.4 \text{ m}$	$\pi(1.5)^2$ $= 7.1 \text{ cm}^2$	7.1×5.8 $= 41.18 \text{ cm}^3$	9.4×5.8 $= 54.52 \text{ cm}^2$

FULL MARKS FOR EACH

Now that you have calculated the volume of each cylinder, put them in order from smallest to largest

Least

Most

1 mark!

depends

on their answers